# Web2Music: Page-Conditioned Music Generation with a Disclosure-Aware Classifier Cascade


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***Abstract*—Text-based music generation models are capable of generating music in accordance with any text description. Nevertheless, a browser extension that generates music relevant to the page being viewed by the user needs to obtain the description from the webpage, to keep the user's browsing history from leaving the device, and to hide the generator that runs slower than the browsing process. We propose Web2Music, a Manifest V3 browser extension that tackles all three challenges. We evaluate the system on 105 live websites, 255 pages of a labeled dataset, 144 generation queries, and 141 generated clips. The page descriptor has a latency of 86 milliseconds per page, excluding the embedding inference. The disclosure-aware classifier cascade that consists of keyword, zero-shot natural language inference, and large language model with an intermediate stage that operates on-device achieves the same macro-averaged F1 score of the keyword-to-large-language-model baseline (0.689) but has 38% fewer pages sent off-device. With the local gate, we can trade accuracy for less off-device disclosure, resulting in a fully local version with a macro-averaged F1 score of 0.594. Fallback-then-swap playback ensures that the audio becomes perceptible 36 milliseconds after the mood commitment, regardless of how long it takes to generate, where, in the case of the Apple M4 machine, generation takes a median of 72 seconds for a cold 28-second segment. The choice of the loop point using a measured seam and not the one based on similarity in chroma lowers the percentage of seams that have more than 1 decibel jump in energy from 74% to 12%.**



***Index Terms*—generative audio, context-aware computing, browser extensions, zero-shot classification, model cascades, on-device inference, ambient interfaces**


## I. Introduction

Background music is chosen once, for a session, and then ignored: a lo-fi playlist runs for four hours across a tax form, a horror wiki and a birthday message without registering that anything changed. Making it context-aware has meant recommending [11] from a library that anticipates every context. Text-conditioned generative audio models remove that requirement (§II): once a description can be derived from what the user is doing, music can be composed for it.

Web2Music does this for the web, and doing it well raises three engineering problems, none about audio quality. **The prompt must be derived:** a page is a DOM of navigation chrome, advertisements, and colour, not a prompt. **Classification is a privacy decision:** the most accurate classifier is a hosted LLM, and calling one on every page would turn a music player into a browsing-history channel. **Generation is slower than browsing:** a cold 28 s clip takes 72 s on an Apple M4 and several minutes on a CPU laptop, against a page load of about two seconds, so the interaction must make generation time invisible.

Web2Music answers each problem with a component we measure. First, a four-signal page descriptor (text, sentence embedding, colour, behaviour) that costs 86 ms per page excluding embedding inference, 4% of navigation time. Second, a disclosure-aware three-stage cascade (keyword $\rightarrow$ zero-shot $\rightarrow$ LLM) that treats the share of pages sent off-device as a

measured, tunable quantity: with an on-device middle stage it matches the keyword-to-LLM baseline's macro-F1 while disclosing 38% fewer pages, and moving that stage's gate sets the trade-off anywhere down to a fully local configuration. Third, fallback-then-swap playback, which makes audio audible 36 ms after mood commitment regardless of generation time, and loop synthesis that turns a one-shot generator into gapless loops; choosing the cut point by measured seam reduces the share of seams with an energy jump above 1 dB from 74% to 12%. Beyond this system, a hosted middle stage only changes which provider sees a page, so what matters is whether a local model is accurate enough; here a quantised one is.

## II. Related Work

MusicGen [20] uses a single autoregressive transformer across four EnCodec [21] codebooks with a delay switching between the codebooks so that a single model generates each stream rather than a chain of models. It uses a frozen T5 encoder to condition on text, optionally on melody, and provides checkpoints ranging from 300M to 3.3B parameters. The smallest model can be run on consumer devices. MusicLM [22] generates hierarchically from semantic tokens to the SoundStream acoustic tokens conditioned on the MuLan embeddings of music and text. These generated tokens remain coherent for many minutes. However, model parameters are not made available. AudioLDM [23] uses CLAP embeddings to perform latent diffusion, which allows training on raw audio. Stable Audio [12] uses timing embeddings so a single model can generate audio of arbitrary length. Both models trade autoregressive generation for a complexity that grows with the number of sampling steps. Text-conditioned environmental sounds are generated by AudioGen [24]. All of these models generate a fixed-length segment of audio where the end of the generated audio is not connected to the start.

LoopGen [1] creates a generative model that can be used to generate loops without retraining. LoopGen adapts MAGNeT, a masked non-autoregressive generative model, to generate its own token sequence in a loop. The end of the sequence is generated with the context of the beginning of the sequence, and the join between the end and the beginning is generated rather than later stitched on. While LoopGen leverages the loopable generation from MAGNeT, the loopable generation does not easily translate to autoregressive models. Instead, a loop can be cut after generation at a structural position identified by self-supervised multi-level representations [13] or barwise self-similarity [25]. AffectMachinePop [2] is an expert system which can generate pop music in real time from a desired valence–arousal trajectory [3] without the use of a GPU. This allows for direct manipulation of the perceived affect in the generated music. However, the desired affect must be provided externally.

There have been three relatively recent approaches that create from user context. MetaBGM [14] uses an LLM to convert game and interactive actions into musical descriptions which are then converted into music by a text-to-music system. The music is maintained across scene changes, but the full state is sent to a cloud LLM on each request. Discussion Jockey 2 [4] transcribes speech from an online meeting and then converts it to prompts for MusicGen following the discussion. A user study found a relaxation effect but the full transcript is sent. Context-AI Tunes [5] uses a combination of a user reported stress level and a picture of their current context to create a prompt for relaxation music. The system was tested on 26 users but has active input and the image exits the device. Casper [6] is a browser extension that cleanses user input prior to sending it to a cloud LLM. It uses a local cascade of identifier rules, NER for locations and names and then an LLM to identify sensitive information. Casper cleanses rather than creates and has no exposure rate at the corpus level.

FrugalGPT [15] created the confidence-gated cascade. Queries are attempted using increasingly cheaper LLMs and a learned scorer makes a decision to either use the response or proceed to the next LLM. It was shown that performance comparable to GPT-4 can be achieved with up to 98% less inference cost. Hybrid LLM [16] attempts each query with the LLM predicted to be most appropriate for the query based on difficulty. Up to 40% fewer large models are used while maintaining output quality. Teacher-student caching [26] uses an online small model trained on outputs from an LLM to reduce the number of times the LLM is queried. Agreement-based cascading [7] only progresses to the next model when a group of low cost models do not agree. These systems focus on reducing cost and latency rather than the content seen by the expensive model. WeInfer [8] demonstrates the viability of running an LLM within the browser using WebGPU.

## III. Research Gaps and Problem Statement

### A. Gaps

Prior work leaves three major research gaps. First, system-level properties have gone unmeasured: evaluation ends at a system's output, typically audio quality. Second, cascades have been optimized for cost efficiency rather than disclosure; no existing cascade permits adjustable exposure. Third, the paucity of generators that produce loops and the missing seams at post-processing have gone unmeasured. Web2Music seeks to close these gaps by systematically deriving music description from any web page passively and attacking all three at once: it enables the measurement of the entire pipeline from the DOM to audio, treating exposure as an adjustable rate, and ensuring all loops are closed with a measurable seam.

### B. Problem statement

For a rendered page $p$, the ambient audio $a(t)$ has four properties. **Latency:** audio begins after a set delay that depends on the mood. **Disclosure:** text that exits the device is measured; sensitive pages produce none. **Continuity:** clips loop without gaps. **Restraint:** pages with sensitive categories produce no audio. The zero-shot and LLM providers, and the key proxy in front of them, receive only the page title, two-sentence description, and top keywords. The generation service receives

only mood, time of day, and tempo; local generation removes that channel entirely. Sensitive pages send nothing.

## IV. System Architecture

Web2Music is a Manifest V3 [17] Chrome extension plus one HTTP generation service. Fig. 1 follows one page from DOM to speaker in nine steps and marks the two persistent stores: an IndexedDB vector store on the device and a Postgres cache index behind the generation service.

### A. Page descriptor

One function assembles the descriptor from four signals: text; a 384-dimensional MiniLM sentence embedding computed in-browser; colour, an area-weighted HSL histogram over up to 800 elements; and behaviour, scroll and cursor velocity; plus a readability score. The embedding is matched by cosine similarity against an IndexedDB store of prior pages to flag revisits, capped at 2,000 entries.

### B. Classifier cascade

The classifier is used to map descriptors to 13 categories and 11 moods, then maps them respectively to the music profile and prompt. **Keyword stage:** at the keyword level, the keyword map is used to find the category, while moods take the keyword score, colour, and behaviour biases together into account [9]. The cascade also stops at this stage, at zero cost, when confidence is $\geq 0.5$. **Zero-shot stage:** the text is scored versus the thirteen category hypotheses using natural-language inference [27]. Once a label's score attains 0.45 and is at least 0.10 larger than the subsequent highest score, the stage will take a label; in any other case, it abstains [7]. The stage has two interchangeable back ends: an online hosted bart-large-mnli endpoint, and a quantised ONNX nli-deberta-v3-xsmall model that runs in the offscreen document without sending any data. **LLM stage:** the gpt-oss-20b model running on Groq is used, with the temperature set to 0. An eight-second timeout limit is implemented; in case of timeout, the decision would be made on the basis of keywords. The chosen mood will stay the same for five seconds in the confidence period before the musical transition to avoid the perceivable fluctuations due to user scrolling. The externally hosted stages are available via a proxy that holds the API keys instead of embedding the credentials into the extension. Every stage's output is logged, so any configuration can be replayed offline (§V-D).

**From mood to prompt.** A committed mood picks a particular row from a per-mood table, including parameters like tempo span, keys, timbre, instruments, reverb, and style. Tempo is calculated using interpolation according to the energy parameter in the chosen range and then tuned with the help of one of nine page-type tuners and seven time-of-day tuners. Key is determined by valence, while darker pages are given more reverb. This profile fills the prompt template ending with loop-based directives (without vocals, without percussion on the loop point).

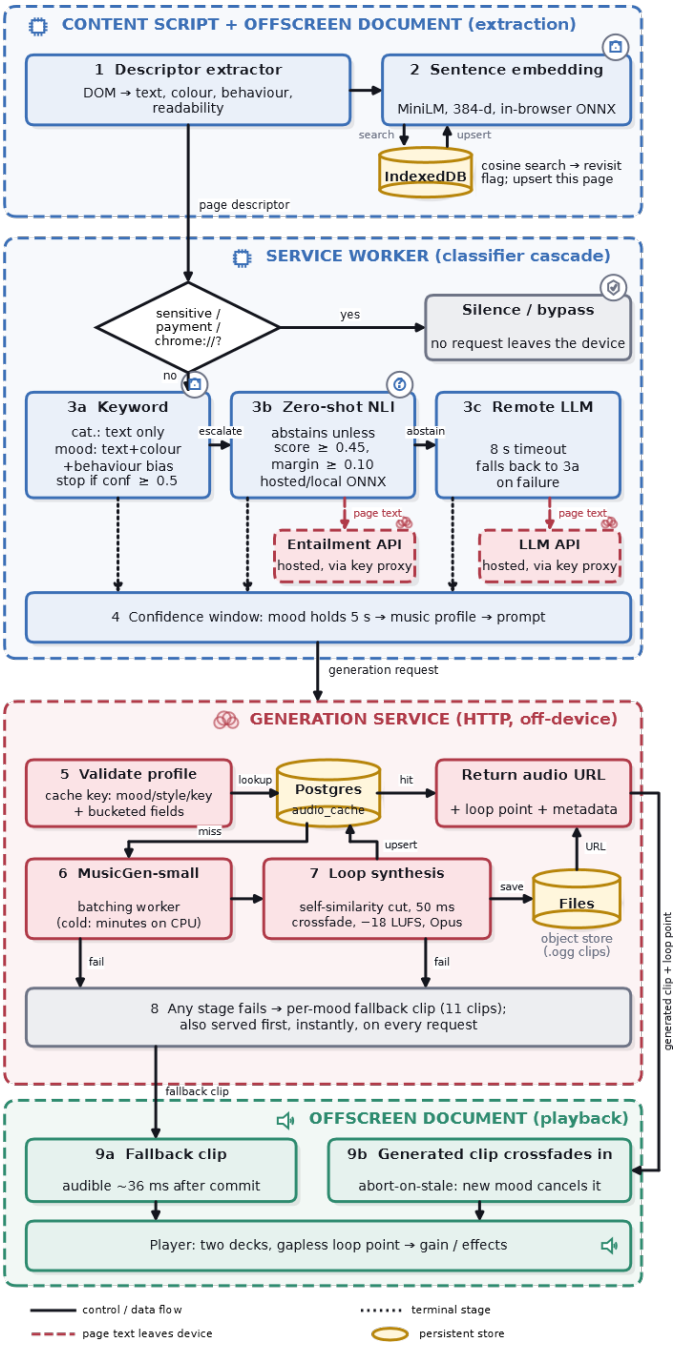



Fig. 1. System architecture of Web2Music.

### C. Generation, loop synthesis and caching

The service verifies the query and queries the Postgres cache index (steps 5–7), where the index is exact on the mood, style, and key but also on the energy, valence, arousal, tempo, and duration buckets. If the match was successful, the URL and loop point are returned. In case of a miss, MusicGen-small (300 million parameters, with default sampling) is run on one batching worker. The worker batches up to four queries within a 150 ms time window, and it generates 50 tokens per second of requested audio that is then processed into an audio clip with index upserting. In case of failure during any step, the

system uses a pre-generated audio clip for the particular mood.

**Loop synthesis.** Chroma descriptors are employed to create a self-similarity curve [25] for potential loop segments, which have a minimum duration of 3 seconds. The best candidate is aligned to a bar boundary, cut there, and then crossfaded over a period of 50 ms using an equal power crossfade. The loudness normalization is performed to $-18$ LUFS [28], after which the clip is exported as an Ogg/Opus file, which, unlike MP3, adds no padding at every repeat. A measured seam selector, as discussed in §V-F, instead examines the same set of candidates and picks the one whose crossfaded seam has the lowest energy and spectral discontinuity.

### D. Playback and restraint

Playback is embedded inside an off-screen document. Looping is performed by two decks using the raw audio buffer inputs alternatively and then applying three gain stages (user volume control, tab reduction, and mood fade), where all the gain stages multiply and not overwrite each other. The fallback-then-swap strategy involves accessing the fallback clip based on mood and the next generation simultaneously, playing back the fallback clip and fading in the new generation based on the equal power curve of the seam for 3 seconds; the previous generation is killed when there is a new mood. **Restraint:** sensitively flagged pages do not have any audio playbacks and are skipped before the start of either of the two hosted stages. When the current tab is generating audio, then the player is muted. On the other hand, background tabs playing audio decrease the gain to 0.1 of the player's volume. URLs of payment sites, banking websites, and internal browser URLs are exempted from the process. The system maintains up to 2,000 telemetry events in SHA-256 hashed form on a local ring buffer, which are never sent out unless exported.

## V. Results and Discussion

### A. Evaluation design

We report a systems benchmark, a cascade ablation, a loop quality study, and a restraint audit. Percentiles are nearest-rank. **Corpus.** 255 real pages in seven slices (general, vocabulary mismatch, non-English, mixed-genre, sensitive, near-miss, bypass), 252 of them category-labelled. **Adversarial slice.** 47 hand-written pages (29 sensitive, 18 not) built from the detector's own term lists to target its failure modes. **Labels.** Each page carries the category it was collected for; three of the authors then labelled every page independently, and where their majority resolves (208 pages) it agrees with that label on 85%. **Clips.** The loop study uses the 141 clips in the service's store, 130 generated and 11 per-mood fallbacks. The systems benchmark ran on a machine with an Intel i7-1255U and 16 GB of memory using CPU-only PyTorch; the cold-generation, duration, and prompt studies ran on an Apple M4 through PyTorch's MPS backend.

### B. Annotation agreement

Over 255 pages, Krippendorff's $\alpha$ [18] is 0.635 for category, 0.419 for binary sensitivity and 0.195 for mood. Category agreement approaches but does not reach the 0.667 reliability threshold, so we score the cascade on category and read its results as tentative. Mood agreement stays low even after coarsening to valence–arousal quadrants [3] ($\alpha = 0.121$) or NRC VAD norms [10] ($\alpha = 0.300$): the mood of a page is a matter of taste. Web2Music therefore treats mood as a stylistic mapping from page signals rather than a classification target.

TABLE I
Cascade configuration results.

| | Configuration | Acc. | Macro-F1 [95% CI] | LLM | Off-dev. |
|---|---|---|---|---|---|
| A1 | keyword only | .190 | .315 [.25, .37] | 0 | 0 |
| A2 | hosted zero-shot only | .556 | .595 [.53, .65] | 0 | .976 |
| A3 | LLM only | .694 | .694 [.64, .74] | 1 | 1 |
| A4 | keyword → LLM | .679 | .689 [.63, .74] | .757 | .757 |
| A5 | keyword → hosted ZS → LLM | .643 | .678 [.62, .73] | .220 | .757 |
| A6 | keyword → local ZS | .433 | .500 [.44, .55] | 0 | 0 |
| A6b | keyword → local ZS → LLM | .671 | .689 [.63, .74] | .467 | .467 |

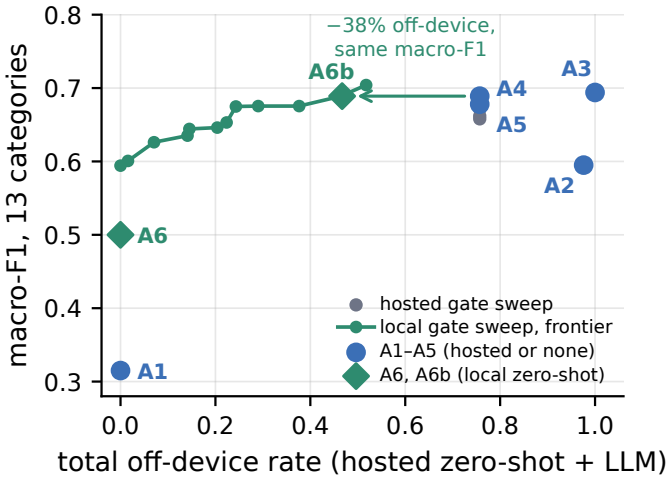



Fig. 2. Macro-F1 versus off-device rate.

### C. Descriptor cost and signal value

On 105 live websites loaded in an automated Chromium browser, descriptor construction takes a median of 86 ms against a 1,989 ms navigation, and boilerplate removal keeps 80.8% of visible text. Embedding inference is stubbed in this benchmark, so 86 ms is the extraction cost excluding the model. Text extraction, the colour histogram, and readability take medians of 11.0, 3.9, and 1.5 ms; most of the remainder, and the whole p95 tail, is the vector-store search, which is idle-scheduled. An offline ablation on 255 pages replays the keyword stage with one signal removed: removing colour changes the mood label on 7.5% of pages, and reducing the descriptor to text alone changes 10.2%, the extra 2.7 points coming from the readability term; both shape about a tenth of mood decisions, which, with no reliable mood ground truth (§V-B), shows influence, not benefit.

### D. Classification and disclosure

Table I and Fig. 2 give the result. The single-stage baselines bracket the design space: keyword-only (A1) discloses nothing at .190 accuracy, and LLM-only (A3) reaches .694 by

TABLE II
LATENCY BY REQUEST PATH.

| Path | HW | n | p50 | p95 |
|---|---|---|---|---|
| Fallback clip served | CPU | 33 | 7 ms | 153 ms |
| Cache hit, end to end | CPU | 93 | 2.09 s | 2.11 s |
| Cache check, pooled connection | M4 | 30 | 8 ms | 41 ms |
| Cold generation, 10–20 s clip | CPU | 3 | 139 s | – |
| Cold generation, 28 s clip | CPU | 12 | >300 s | – |
| Cold generation, 28 s clip, one mood | M4 | 30 | 72.1 s | 105.9 s |
| Cold generation, 28 s clip, 11 moods | M4 | 110 | 72.2 s | 119.4 s |
| Navigation to audible (client) | CPU | 54 | 6.86 s | 11.28 s |

disclosing every page. Keyword-to-LLM (A4) comes within .005 of A3's macro-F1 (paired 95% CI $[-.041, +.033]$) while keeping 24% of pages on the device. Every macro-F1 interval is about $\pm.05$ wide, so smaller differences are not resolved by this corpus.

A local middle stage is what makes the cascade a privacy mechanism. With the on-device zero-shot model, A6b matches A4's macro-F1 (.689; paired difference .000, 95% CI $[-.044, +.042]$) while the fraction of pages leaving the device falls from .757 to .467, a 38% reduction. The local model answers in a median of 390 ms against 1.74 s for the hosted one.

The local gate is the disclosure dial. Replaying A6b with the local stage's gate moved traces the green frontier in Fig. 2. Lowering the score threshold from 0.45 to 0.30 cuts off-device from .467 to .290 at macro-F1 .675, and opening the gate entirely keeps every page on the device at .594, level with the hosted zero-shot model alone (A2, .595); A6 at the default gate reaches only .500 because declined pages fall to a default category. The gate cells are scored on the corpus they are chosen from, so they locate the trade-off rather than tune it.

A hosted middle stage only redistributes disclosure. A5 cuts LLM-specific exposure from .757 to .220, but every page it keeps from the LLM reaches the hosted zero-shot endpoint instead, so total off-device stays at .757, as the cascade's structure dictates. A twelve-cell sweep of the hosted gate shows it: macro-F1 stays within .658–.685 and LLM exposure moves over a $5.4\times$ range (.106–.576), while total off-device holds at .756–.757 in every cell. The hosted gate chooses which provider sees a page; only a local backend changes whether one does.

### E. Generation latency and fallback-then-swap

Table II and Fig. 3 show the routes a request can take. The fallback clip is served in 7 ms, so the audible start is never delayed by the generation process. On the Apple M4, all 140 cold generations of 28 seconds complete, with a median of 72 seconds and a real-time factor of 1.8–2.3 across durations; the generation process itself takes up 71.5 seconds of that median, and every other stage takes under a second. The CPU laptop is approximately four times slower (real-time factor 8.2–10.1): its three completed cold generations took a median of 139 seconds, and 12 out of 144 requests, all cold 28-second misses queued one after another on a single worker, exceeded the 300-second client timeout; the fallback clip played the whole time.

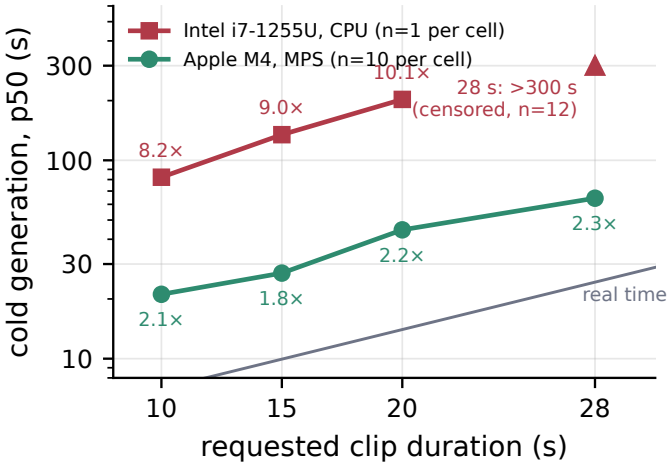



Fig. 3. Generation time versus clip duration.

TABLE III
LOOP-SEAM QUALITY BY CUT METHOD.

| Cut point, crossfade | $\lvert\Delta E\rvert$ p50 | $\lvert\Delta E\rvert$ p95 | >1 dB | $\lvert\Delta c\rvert$ p50 |
|---|---|---|---|---|
| Chroma similarity, 50 ms (default) | 2.44 dB | 10.29 dB | 74% | 405 Hz |
| Chroma similarity, 100 ms | 2.03 dB | 7.87 dB | 81% | 314 Hz |
| Chroma similarity, 250 ms | 2.18 dB | 7.41 dB | 71% | 190 Hz |
| Chroma similarity, 500 ms | 1.82 dB | 7.56 dB | 67% | 133 Hz |
| Measured seam, 50 ms | 0.14 dB | 1.54 dB | 12% | 288 Hz |

Cache hits on the CPU laptop took 2.09 seconds, almost all of it spent in the cache lookup, and using pooled Postgres connections lowers that lookup to 8 ms.

An automated browser tests the built extension against eleven fixture pages with generation and classification stubbed out: out of 55 page loads, 54 became audible, with a median latency of 6.86 seconds from navigation. The first extraction finishes in 0.86 seconds, and the mood then stays stable throughout the confidence window (median of 6.0 seconds); once the mood commits, the fallback becomes audible in a median of 36 milliseconds (p95: 54 milliseconds), regardless of what the generator is doing. This is the latency property of §III: the remaining latency is the confidence window, which is a design parameter, not model latency. The p95 value (11.28 seconds) is due to one fixture requiring five extraction passes before its mood commits.

### F. Loop quality

Of the 141 delivered clips, all fall within their duration bounds, and loudness lands at $-18.09$ LUFS against the $-18.0$ target. The 50 ms crossfade reduces the median seam energy discontinuity from 3.73 to 2.64 dB and improves 58.9% of seams. Delivered loops keep a median 50.3% of the requested duration; since the loop repeats gaplessly, this sets the loop period rather than the playback length, so a 28 s request repeats about every 14 s.

Table III and Fig. 4 compare the two levers on the same 113 clips. Increasing the crossfade tenfold, from 50 to 500 ms, lowers the median energy jump by only 25%, and 67% of seams remain above 1 dB. The measured-seam selector, which

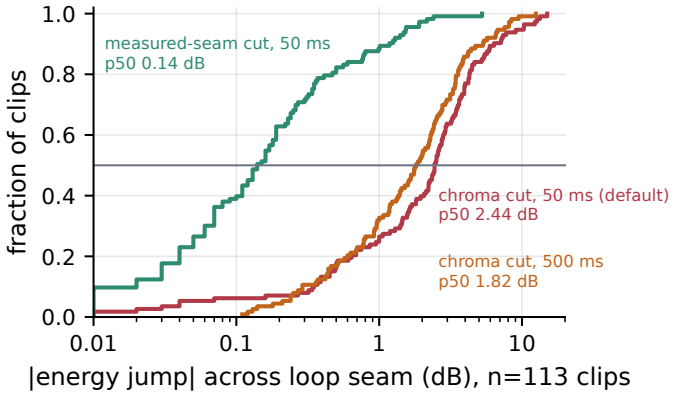



Fig. 4. Energy jump across loop seams.

TABLE IV
PROMPT COMPARISON.

| Prompt | Chars | $\lvert\Delta E\rvert$ | $\lvert\Delta c\rvert$ | Retention |
|---|---|---|---|---|
| P1 bare mood word | 5 | 3.75 dB | 565 Hz | .527 |
| P2 service template | 175 | 2.63 dB | 509 Hz | .444 |
| P3 engineered (default) | 312 | 4.35 dB | 316 Hz | .393 |

keeps the 50 ms crossfade and chooses among the detector's own candidate points, brings the share of seams above 1 dB down to 12% and above 3 dB to 1% (from 74% and 36%), and the median drops from 2.44 to 0.14 dB. It makes the seam better on 102 of 113 clips, worse on two, and preserves internal duration retention (.695 vs. .690). The selector minimises this same energy jump together with a small spectral-centroid term, so these numbers show how far the candidates let the jump fall, not that listeners hear it; the 500 ms crossfade still gives the smallest centroid jump.

### G. Prompt ablation

The default prompt is an engineered description of 312 characters. On the Apple M4, we tested it against the bare mood word and the service's 175-character template, keeping mood, duration and seed constant (Table IV).

Both shorter prompts outperform the engineered one on energy seam and duration retention, the engineered prompt wins only on spectral-centroid seam, and all three take the same time to generate (18.2–18.3 s). The template yields the smallest energy jump, and the mood word yields the longest loops. Prompt engineering therefore brings no improvement in loop quality here; with 15 generations per prompt and no statistical testing, this is an indication rather than a ranking, and whether listeners prefer the engineered prompt's timbre remains a question for the listening study.

### H. Restraint

All fourteen executable restraint policies pass their tests: silence on sensitive pages [19], muting and ducking around existing audio, bypass of payment and browser-internal pages, hashed-URL telemetry, and seven unit cases. The adversarial slice then attacks the sensitivity predicate.

TABLE V
SENSITIVE-PAGE DETECTION.

| Detector | Missed | False alarms | Non-English caught |
|---|---|---|---|
| Term lists (default) | 19/29 | 3/18 | 0/4 |
| Term lists + entailment (hosted) | 0/29 | 11/18 | 4/4 |

The term lists are precise, producing only three false positives, all near-miss academic pages such as a grief-counselling degree. However, a white-box slice built from the lists' own vocabulary evades them on 19 of 29 sensitive pages, through out-of-list terms (7), euphemisms (4), non-English text (4), and single ambiguous terms (4). Re-scoring with the entailment model detects all 29, including the four non-English pages, with scores between 0.83 and 0.92 (Table V). This run used the hosted bart-large-mnli backend, which receives every page without a hard-severe term; in this configuration, the check itself therefore discloses the pages it is meant to protect. The disclosure property holds only with the on-device backend, which has not yet been run on this slice. The check also overtriggers: its eleven false positives include all six plainly benign pages, such as a Rust tutorial, because the prompt offers only two hypotheses, crisis or reference discussion, without an "unrelated" option, so the check stays optional until it is re-measured.

## VI. Conclusion

The Web2Music system creates background music that is suitable for the context of the current page that the user is reading. The pipeline steps of the system are assessed based on the characteristics outlined in Section III. Running the intermediate step of the pipeline on-device keeps the keyword-to-LLM macro-F1 score intact and cuts down off-device page transfers by 38%, whereas the relevant gate controls the level of reduction of further information leakage. Playback of audio starts 36 milliseconds after the mood decision, irrespective of the duration of generation. Choosing the loop point by measured seam also removes most of the energy jumps at the loop.

## VII. Limitations and Future Work

The following stage will be a listening experiment (within-subjects design) pending ethics clearance to test page-matched audio versus shuffled audio in terms of forced-choice fit, Likert rating, dwell times, skip rate and mute rate. Also, the experiment will find out if cleaner seams can be heard and if the loop period of 14 seconds can cause fatigue. There are three other steps expected in the engineering process: (1) making A6b and the measured seam selector the default settings, (2) performing the entailment sensitivity test on the on-device back end, employing an "unrelated" hypothesis, and re-measuring the two error rates, and (3) conditioning the generator to create loop-closed content.

The benchmarks do not contain any model inference; the authors are the annotators, and the gate cells are tested based on the corpus from which they were extracted. The next

set of tests will time the embedding model, use independent annotators, and use a held-out corpus. An analogous pipeline could also be used for further continuous contexts such as documents and code editors.

## References


[1] D. Marincione, G. Strano, D. Crisostomi, R. Ribuoli, and E. Rodolà, "LoopGen: Training-free loopable music generation," in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, 2025.

[2] K. R. Agres, A. Dash, P. Chua, and S. K. Ehrlich, "AffectMachinePop: A controllable expert system for real-time pop music generation," presented at AAAI Workshop Artif. Intell. Music, 2025.

[3] J. Kang and D. Herremans, "Are we there yet? A brief survey of music emotion prediction datasets, models and outstanding challenges," *IEEE Trans. Affect. Comput.*, vol. 16, no. 4, pp. 2545–2559, 2025.

[4] H. Suzawa, K. Watanabe, A. Dengel, and S. Ishimaru, "Augmenting online meetings with context-aware real-time music generation," in *Proc. Augmented Humans Int. Conf.*, 2025, pp. 487–490.

[5] X. Wei, Z. Zhang, Z. Yue, and H.-T. Chen, "Context-AI Tunes: Context-aware AI-generated music for stress reduction," in *Artif. Intell. HCI (HCII)*, LNCS, 2025, pp. 330–345.

[6] C. J. Chong, C. Hou, Z. Yao, and S. M. Seyed Talebi, "Casper: Prompt sanitization for protecting user privacy in web-based large language models," in *Proc. IEEE CSCloud*, 2025, pp. 122–133.

[7] S. Kolawole, D. Dennis, A. Talwalkar, and V. Smith, "Agreement-based cascading for efficient inference," *Trans. Mach. Learn. Res.*, 2025.

[8] Z. Chen, Y. Ma, H. Shen, and M. Liu, "WeInfer: Unleashing the power of WebGPU on LLM inference in web browsers," in *Proc. ACM Web Conf.*, 2025, pp. 4264–4273.

[9] D. Jonauskaite and C. Mohr, "Do we feel colours? A systematic review of 128 years of psychological research linking colours and emotions," *Psychon. Bull. Rev.*, vol. 32, no. 4, pp. 1457–1486, 2025.

[10] S. M. Mohammad, "NRC VAD Lexicon v2: Norms for valence, arousal, and dominance for over 55k English terms," arXiv:2503.23547, 2025.

[11] Y. Deldjoo, M. Schedl, and P. Knees, "Content-driven music recommendation: Evolution, state of the art, and challenges," *Comput. Sci. Rev.*, vol. 51, Art. no. 100618, 2024.

[12] Z. Evans, C. J. Carr, J. Taylor, S. H. Hawley, and J. Pons, "Fast timing-conditioned latent audio diffusion," in *Proc. Int. Conf. Mach. Learn.*, 2024.

[13] M. Buisson, B. McFee, S. Essid, and H. C. Crayencour, "Self-supervised learning of multi-level audio representations for music segmentation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 2141–2152, 2024.

[14] H. Liu *et al.*, "MetaBGM: Dynamic soundtrack transformation for continuous multi-scene experiences with ambient awareness and personalization," arXiv:2409.03844, 2024.

[15] L. Chen, M. Zaharia, and J. Zou, "FrugalGPT: How to use large language models while reducing cost and improving performance," *Trans. Mach. Learn. Res.*, 2024.

[16] D. Ding *et al.*, "Hybrid LLM: Cost-efficient and quality-aware query routing," in *Proc. Int. Conf. Learn. Represent.*, 2024.

[17] N. Pantelaios and A. Kapravelos, "Manifest V3 unveiled: Navigating the new era of browser extensions," in *Proc. NDSS Workshop MADWeb*, 2024.

[18] G. Marzi, M. Balzano, and D. Marchiori, "K-Alpha Calculator–Krippendorff's Alpha Calculator: A user-friendly tool for computing Krippendorff's alpha inter-rater reliability coefficient," *MethodsX*, vol. 12, Art. no. 102545, 2024.

[19] H. Qiu, L. Ma, and Z. Lan, "PsyGUARD: An automated system for suicide detection and risk assessment in psychological counseling," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2024, pp. 4581–4607.

[20] J. Copet *et al.*, "Simple and controllable music generation," in *Adv. Neural Inf. Process. Syst.*, 2023.

[21] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, "High fidelity neural audio compression," *Trans. Mach. Learn. Res.*, 2023.

[22] A. Agostinelli *et al.*, "MusicLM: Generating music from text," arXiv:2301.11325, 2023.

[23] H. Liu *et al.*, "AudioLDM: Text-to-audio generation with latent diffusion models," in *Proc. Int. Conf. Mach. Learn.*, 2023.

[24] F. Kreuk *et al.*, "AudioGen: Textually guided audio generation," in *Proc. Int. Conf. Learn. Represent.*, 2023.

[25] A. Marmoret, J. E. Cohen, and F. Bimbot, "Barwise music structure analysis with the correlation block-matching segmentation algorithm," *Trans. Int. Soc. Music Inf. Retrieval*, vol. 6, no. 1, pp. 167–185, 2023.

[26] I. Stogiannidis, S. Vassos, P. Malakasiotis, and I. Androutsopoulos, "Cache me if you can: An online cost-aware teacher-student framework to reduce the calls to large language models," in *Findings Assoc. Comput. Linguist.: EMNLP*, 2023, pp. 14999–15008.

[27] M. Laurer, W. van Atteveldt, A. Casas, and K. Welbers, "Building efficient universal classifiers with natural language inference," arXiv:2312.17543, 2023.

[28] European Broadcasting Union, "Loudness normalisation and permitted maximum level of audio signals," EBU Rec. R 128, ver. 5.0, 2023.